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Imaging of root zone processes using MRI T_1 mapping

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ABSTRACT

Noninvasive imaging in the root soil compartment is mandatory for improving knowledge about root soil interactions and uptake processes which eventually control crop growth and productivity. Here we propose a method of MRI T_1 relaxation mapping to investigate water uptake patterns, and as second example, in combination with neutron tomography (NT), property changes in the rhizosphere.

The first part demonstrates quantification of solute enrichment by advective transport to the roots due to water uptake. This accumulation is counterbalanced by net downward flow and dispersive spreading. One can furthermore discriminate between zones of high accumulation patterns and zones with much less enrichment. This behavior persists over days. The second part presents the novel combination of MRI with neutron tomography to couple static, proton density information of roots and their interface to the surrounding soil with information about the local water dynamics, reflected by NMR relaxation times. The root soil interface of a broad bean plant is characterized by slightly increasing MRI and NT signal intensity but decreasing T_1 relaxation time indicating locally changed soil properties.

1. Introduction

The exploration of root soil interactions, water and solute transport in the root zone is necessary for the understanding how plants try to manage stresses related to water scarcity. Moreover, one needs exact knowledge about the root system architecture and its relationship to the involved root zone and rhizosphere processes [1–5]. These processes are classically investigated by rhizotrons which restrict growth and observation to a very thin 2D layer or one draws conclusions only indirectly by analyzing breakthrough curves and thereby treating the soil root compartment as a black box [6]. This type of macroscopic characterization fails to see the details to unambiguously resolve root zone processes like water uptake, solute transport and changes in soil properties.

The purpose of the present work is to demonstrate the suitability of MRI for root system imaging and T_1 mapping to obtain information about root processes such as solute enrichment and modification of the soil in the rhizosphere. For the latter, MRI is additionally combined with neutron tomography (NT) to investigate separately the influences of water content and changes of relaxation times. Solute transport is controlled by slow flow patterns and therefore we have monitored it by concentration changes of the contrast agent GdDTPA using T_1 mapping [7]. Exudation of mucilage, microbe activity, and presence of root hairs may cause changes in water retention properties around individual roots [3,8]. Neutron tomography is a valuable tool to probe the absolute proton density, i.e moisture [9]. However, it fails to image molecular mobility in the liquid phase. Vice versa the advantage of MRI is its sensitivity to rotational and translational mobility by relaxation time weighted sequences or even by direct relaxation time mapping. Consequently, a combination of both methods is followed for the investigation of the second root zone process.

2. Theory

2.1. T_1 mapping by IR-preparation

The new T_1 mapping approach is an extension of the T_1-weighted spin or gradient echo experiments mostly used in medical MRI. It bases upon a set of inversion recovery prepared spin echo multislice images (IR-SEMS) with several inversion times which are normalized on a reference image with exactly identical settings but omitting the inversion filter. The voxel intensity is given by Eq. (1) (for details see Ref. [7]):

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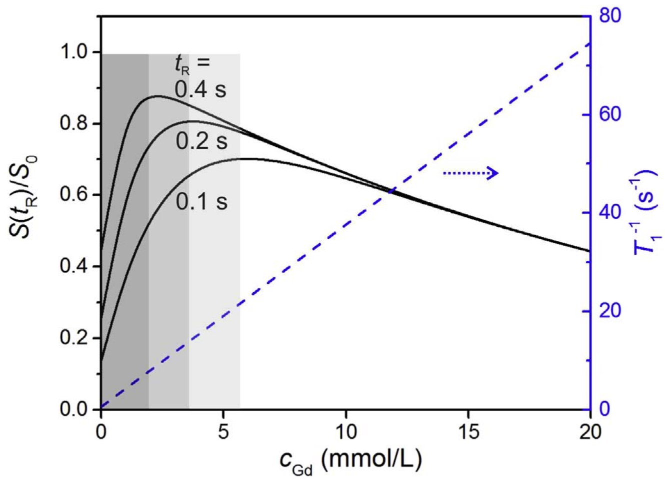

Fig. 1. Comparison of the sensitivity of T_1 weighted (t_R = 0.1, 0.2, and 0.4s) and T_1-mapping pulse sequences on the determination of tracer concentration. The shaded areas indicate the respective accessible concentration ranges with unambiguous concentration determination using T_1 weighted sequences.

$$\frac{S(t_{inv})}{S_{ref}} = \left(1 - 2\exp\left[-\frac{t_{inv}}{T_1}\right]\right) \tag{1}$$

which is used to fit T_1 curves for each voxel. The real part of the signal runs from −1 to +1, and the imaginary part contains only noise, a consequence of precise phasing and the use of a hyperbolic secant adiabatic full passage inversion pulse. The resulting T_1 maps are used to conclude either on changes in soil properties or on contrast agent concentration. The latter is calculated by the linear relationship:

$$\frac{1}{T_1} = \frac{1}{T_{1,bulk}} + \frac{1}{T_{1,surface}} + a_1 c_{Gd} \tag{2}$$

where c_{Gd} is the concentration of $GdDTPA^{2-}$ in the liquid phase of the porous medium, and a_1 the specific relaxivity, independent of the pore size and saturation. The greatest advantage of this approach becomes clear by a comparison with the conventionally used T_1 weighted approach as shown in Fig. 1. There we plot the relative intensity of a T_1 weighted image for a set of repetition times as a function of Gd concentration and compare it with the linear relation from Eq. (2). The approximately linear range becomes wider with shorter repetition time t_R but on the expense of decreasing signal intensity. If one would use T_1 weighted sequences at concentrations higher than a certain threshold beyond which the T_2 decay becomes dominant, the assignment of relative signal intensities is ambiguous. The advantage of T_1 mapping by IR preparation is the linear relation between $1/T_1$ and c_{Gd} so that a limitation of the accessible concentration range depends only on the shortest possible inversion time. The possibility of monitoring larger concentration ranges offers the opportunity to follow and subsequently to describe concentration change processes in their entirety.

3. Experimental

3.1. Plant setup

White lupin *(Lupinus albus)* and broad bean (*Vicia faba*) seeds were germinated on wet paper and implanted three days later into the MRI cuvettes filled with natural medium sand (FH31, Quarzwerke Frechen GmbH, Frechen, Germany). The plant grew under artificial light at 450 μmol/(m^{2}s) and 25 °C in a 12 h/12 h day-night cycle at relative humidity of about 40%.

For the investigation of the root zone process of solute enrichment a *Lupinus albus* seedling was planted in a 50 mm × 150 mm Perspex cylinder filled to 100 mm height with medium sand FH31. It was regularly irrigated with N-P-K nutrient solution (0.14 g/kg N, 0.03 g/kg P, 0.08 g/kg K) to maintain mean water content between 0.15 and 0.25 cm^3/cm^3. After 18 days the systems was flushed with pure tap water to achieve constant high water content and to remove surplus nutrient. Afterwards, MRI experiments were performed daily during a period of 7 days under steady state irrigation with 1 mM $Na_2GdDTPA$ solution.

For the investigation of the soil properties by combination of MRI and NT boron free quartz glass cuvettes with an inner diameter of 27 mm and a height of 100 mm were chosen. *Vicia faba* was grown for 10 days before starting the imaging without additional nutrients at moderate water content between 0.15 and 0.25 cm^3/cm^3.

3.2. MRI

For both root zone processes T_1 maps were obtained using a 1.5 T MRI scanner at the Research Center Jülich (Agilent Technologies) with a 2D spin echo multi slice sequence with inversion recovery preparation (IR-SEMS) including a hyperbolic secant adiabatic full passage (HS-AFP) 180° inversion pulse. The images with five different inversion recovery preparation times t_{inv} were normalized on the reference image, and finally T_1 for each voxel was derived by fitting Eq. (1) to measured $S(t_{inv})/S_{ref}$ for the different t_{inv}. The total time was about 100 min for six scans. The T_1 maps were further used to calculate Gd concentrations for the first root process (section 4.1) using Eq. (2). For more details the reader is referred to [7]. For the second root process T_1 maps were analyzed as described in section 4.2.

In parallel, root soil systems were imaged separately using T_2-weighted 3D fast spin echo imaging sequences on the 1.5 T scanner at the research center Jülich (Agilent Technologies, Varian console) and on a transportable 0.56 T Halbach-type scanner made of permanent magnets (Magritek, Wellington, New Zealand). The total time was 34 min with the stationary 1.5 T scanner and 140 min with the transportable instrument.

3.3. Neutron tomography

The neutron experiments were performed at the Paul Scherrer Institute using the imaging station NEUTRA [10]. Samples were imaged at the middle measure position of the instrument where the collimation ratio L/D is 350 using a 100 μm thick Li-6 ZnS scintillator as detector screen and captured by a Neo sCMOS camera at a resolution of 97 μm/pixel. 500 radiographic projections were taken using an exposure time of 5 s while the sample was rotated stepwise over an angular range of 360°. Data processing comprised darkfield and flatfield correction, normalization to the intensity values of a sample-free region, and removal of white spots in the radiograms by the “remove outlier” filter of ImageJ. After that the 3D information of the sample was calculated using the reconstruction software Octopus. 3D graphics were rendered using the software package VGStudioMax.

4. Results and discussion

4.1. First root zone process: solute transport

Solute enrichment was monitored during continuous irrigation with 1 mM $Na_2GdDTPA$ solution under steady state, i.e. the transpiration volume to efflux was 4:1. Before each imaging of the tracer distribution the root system architecture was measured using a T_2-weighted 3D fast spin echo sequence with a resolution of 0.23 mm × 0.23 mm × 0.55 mm. The images help to discriminate the position of the roots from the soil in the T_1 and Gd maps with their lower resolution in vertical direction. The root system was already well developed showing a taproot growing through the entire cuvette (Fig. 2a). To demonstrate the solute transport an axial slice at 15 mm below the surface is chosen in Fig. 2b and d. The exemplary T_1 map of

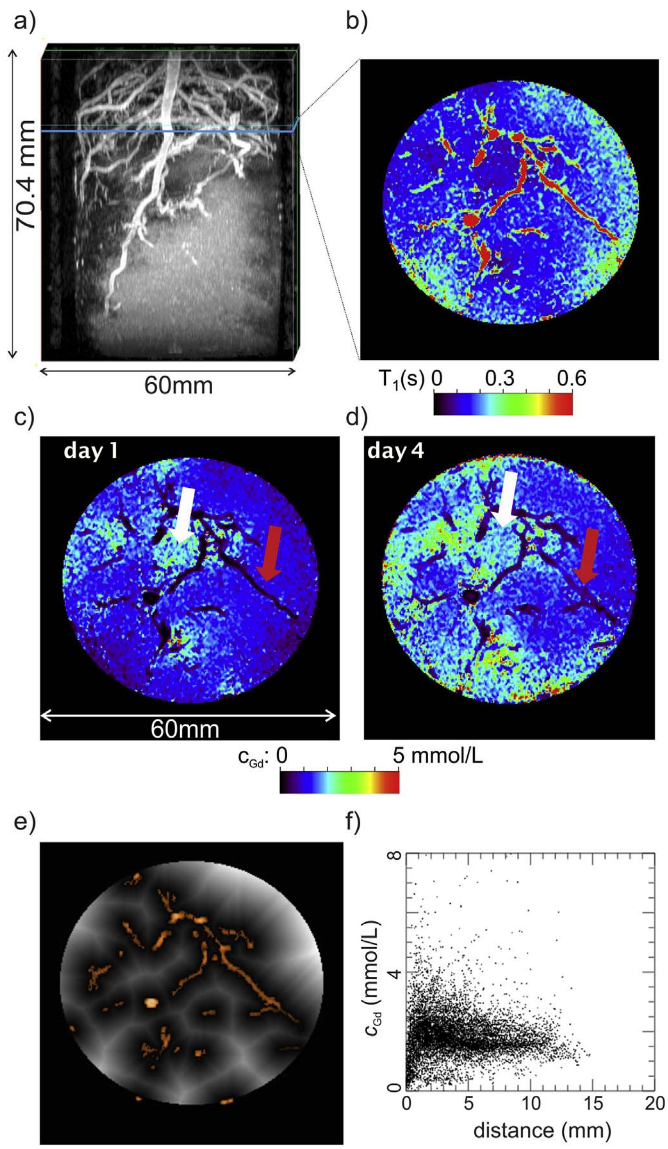

Fig. 2. Solute enrichment in the root zone of a lupin plant. a) Maximum intensity projection of the root system architecture (RSA) imaged by a 3D fast spin echo sequence with a resolution of 0.23 mm × 0.23 mm × 0.55 mm. The blue line indicates the selected axial slice 24 mm below soil surface to follow the solute enrichment process plotted in b) to d). b) T_1 map of the selected slice. Please note that the scale runs to a maximum of 0.6 s so that the longer relaxation times inside the root are also represented by the red color. c) and d) Gd concentration maps of day 1 and day 4 after start of the irrigation show the enrichment of $GdDTPA^{2-}$ around several roots (green arrow) whereas some roots remain mostly inactive (red arrow). The resolution is 0.23 mm × 0.23 mm with a slice thickness of 2.2 mm e) Euklidean distance map of the chosen slice. Overlaid in orange color is the root system in the chosen slice which was segmented by region growing in IDL. f) Correlation plot of Gd concentration as function of the distance to closest root. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

day 1 shows two different time ranges: In the soil compartment T_1 is mostly shorter than 0.2 s due to presence of GdDTPA, whereas the roots are characterized by considerably longer T_1 above 0.4 s (Fig. 2b). The corresponding Gd maps (Fig. 2c and d) visualize the accumulation of Gd around several roots already one day after start of the irrigation. Important is that this accumulation proceeds heterogeneously. That means the root system can be divided into active (green arrow) and less active parts (red arrow). This state persists for longer time since also after four days concentrations increased in similar areas. To illustrate the enrichment process further, we correlated the tracer concentration in individual voxels with their distance to the closest root. First, the roots were segmented by the region grow procedure implemented in IDL. Then a distance map was calculated using Euklidean distance for each non-root voxel, see Fig. 2e. Finally, the concentrations in the individual voxels were plotted against the voxel distance resulting in the correlation plot shown in Fig. 2f. An average enrichment zone at about 2 mm around the roots is clearly visible but enhanced concentrations persist up to 12 mm outside the roots. It is worth mentioning that we speak of areas of accumulation around the roots since there is interplay between advective enrichment connected to local flow towards the roots and dispersive spreading by the net vertical flow. The entire enrichment process was monitored over a time range of seven days, and from images published in Ref. [7] it is known that the concentration enrichment at the roots exceed 10 mmol/L. This example underlines our argumentation concerning the benefits of IR preparation for T_1 mapping. This method refinement allows monitoring a large concentration range, a prerequisite for understanding the entire process.

4.2. Second root zone process: changed soil properties

Changed soil properties may occur at the interface of bulk soil and root and they are influenced by exudation of mucilage, microbial growth and root hairs. Generally, local NMR relaxation times are controlled by both local water content and the dynamics of the liquid influenced by chemical-physical properties of the porous medium like pore size and polymer content. On the other hand, neutron tomography is sensitive for the absolute proton density, i.e. water content. Therefore a combination of neutron tomography (NT) and MRI would make it possible to differentiate if zones with reduced relaxation times are caused by locally decreased water content revealed as by NT or by changes in the local properties of the pore space, which will enhance NMR relaxation and becomes visible in relaxation time maps. Such changes are best discussed in detail by analyzing cross section profiles through an exemplary lateral root and the neighboring soil region from MRI and NT images (Fig. 3a and c), averaged in an area indicated by red rectangles. The averaged cross section profile of the MRI image (green dashed curve in Fig. 3e) measured at an average water content of $0.34\ cm^3/cm^3$ on day 0 shows higher signal intensity inside the roots due to the high water content of root. It decreases to the same value on both sides of the root to a value identical to the intensity in the soil, i.e. the average soil water content. The shape of the profile observed two days later (black dashed curve in Fig. 3e) with the transportable MRI scanner shows no deviations in the root and confirms that the shaded part of the profile corresponds to the root indicating its thickness of 1.6 mm. This work allows for the first time to compare MRI and NT images (Fig. 3b and c) of an identical plant measured directly after each other. The corresponding NT profile is included in Fig. 3e as red curve. It matches the MRI profiles perfectly with only minor differences due to the smaller slice thickness of the NT image (0.097 mm in NT vs. 0.63 mm in MRI). However, both profiles reflect the average water content change during two days.

In the last step the MRI and NT profiles are supplemented by the profile from the T_1 map (Fig. 3d) measured under nearly saturated conditions ($\theta = 0.34\ cm^3/cm^3$) on day 0 (blue curve in Fig. 3e). The grey shaded area indicates that the relaxation behavior inside the root reflects the transition from the anatomical regions of the cortex to the stele ($T_1 = 1.2$ s) and back to the cortex ($T_1 = 0.9$ s) [11,12]. On both sides of the root the relaxation behavior is symmetric and shows continuous increase of T_1 to its value of 1.4 s characteristic for the saturated bulk soil [13]. This zone with a thickness of 1 mm extends much larger than the range where T_1 is expected to be accelerated by diffusional exchange between bulk solution and the interface, which is about 0.1 mm [14]. Moreover, this zone corresponds approximately to the extension of the rhizosheath, i.e. the direct surrounding of the root in the root soil interface. It should be noted that in this zone the signal intensities of MRI as well as of NT increased towards the root, whereas T_1 decreased. However, we interpret the change in the relaxation times

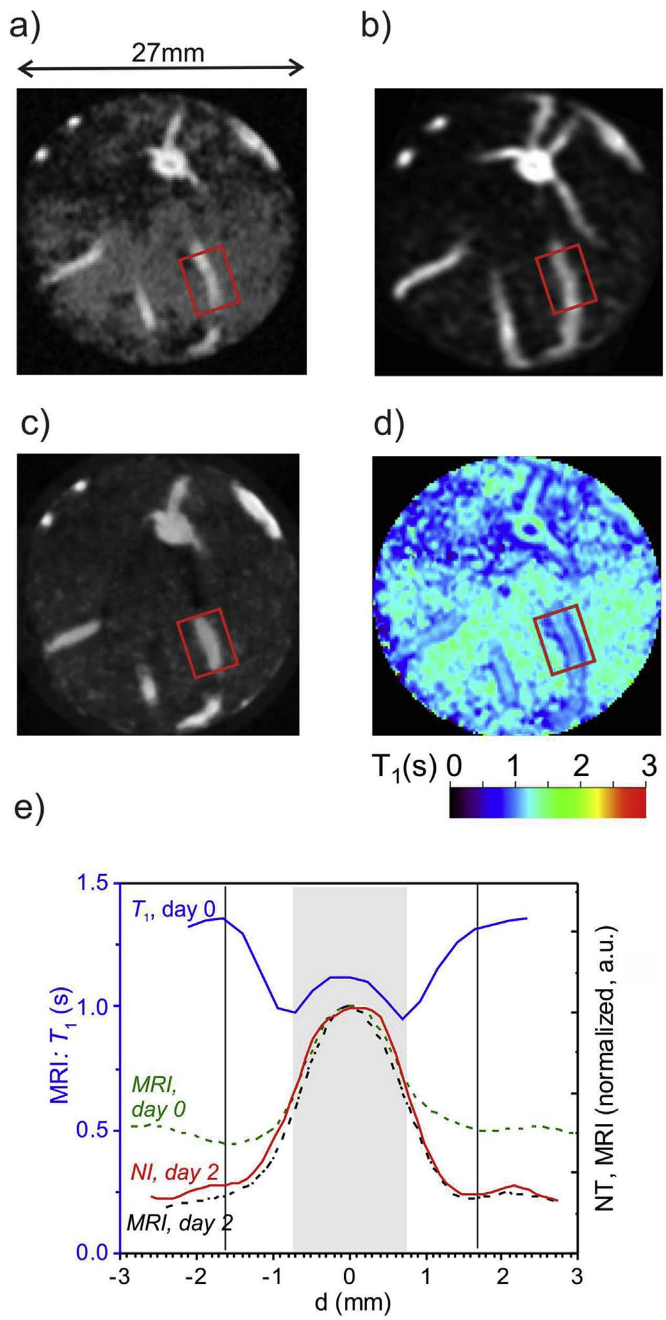

Fig. 3. Root system of a broad bean plant in sand: exemplary selected axial slices at identical positions and orientation after co-registration. a) MRI-SEMS image on day 0 with an in-plane resolution of 0.23 mm × 0.23 mm and slice thickness of 2.2 mm b) MRI-RARE image on day 2 measured with the transportable scanner with an in-plane resolution of 0.23 mm × 0.23 mm and 0.63 mm in vertical direction after zero filling. c) Neutron image on day 2 with an isotropic resolution of 0.097 mm d) T_1 relaxation time map with identical resolution as the SEMS image in a). e) Profiles of T_1 (blue curve, left ordinate), and normalized image intensities (right ordinate) through the lateral root and its vicinity. The profiles are averaged along a ca. 5 mm long section indicated by the red rectangles in a) to d). All profiles are normalized to the respective maximum intensities. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

as changes in the material composition caused by mucilage exudation and root hair growth in the rhizosheath.

5. Conclusions

In conclusion, MRI T_1 mapping described in this paper is a valuable tool to investigate root zone processes. The greatest advantages of IR prepared SEMS with normalization on a reference image are the compensation of T_2 decay and water content inhomogeneities, the avoidance of cross talk artefacts, and the doubled signal amplitude compared to t_R weighted images. One should note that also a normalization of t_R weighted images on a reference image would compensate for T_2 relaxation. But these images still have less signal intensity and they are prone for cross-talk artifacts. This is avoided in IR prepared images by setting $t_R > 5 \times T_1$. However, these advantages are paid with a significant longer overall measuring time. With this new procedure one can monitor solute transport in high resolution by relaxation time mapping, from which concentration changes could be quantitatively derived over a large concentration range. The second root zone process of changed soil properties was characterized in combination with neutron tomography and MRI imaging using a transportable Halbach system. Over a small distance of 1 mm T_1 decreased from its value characteristic for saturated bulk soil, whereas MRI and NT signal intensities slightly increased. From this behavior one can conclude that the rhizosheath in the root soil interface has different material properties, possibly caused by root exudation and root hair growth.

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